Seasonal Shoreline Retreat on Gravel Beaches: A Two-Year Survey

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Abstract

We surveyed five gravel beaches along the Eastvale coast every month between October 2024 and September 2026. Mean shoreline retreat was 3.6 m over the two years, but individual sites ranged from 1.1 m to 6.8 m. Retreat was strongly seasonal and was concentrated in months with more than two storm events.

1 Introduction

Gravel beaches protect many small coastal communities, yet they are monitored far less often than sandy shores. Their steep profiles respond quickly to waves, and a single storm can move material by several metres.

Earlier surveys on this coast were made once a year and could not separate storm damage from slow trends. Retreat peaked in the winter of 2025.[1]

Managers need to know whether retreat is steady or driven by a few events. The answer decides how often a beach must be surveyed and where defences should be built.

This paper reports the first two years of a monthly survey designed to fill that gap. It describes the field method, the results for five sites and the lessons for coastal managers.

2 Methods

Each site was visited on the first Monday of the month at low tide. We recorded the shoreline with a handheld receiver following three steps.

(i) walk the mean high-water line from the north marker to the south marker;

(ii) record one position every two metres along the line;

(iii) repeat the walk if the receiver reports an error above 0.1 m.

[1] Winter is defined as December to February.

2.1 Processing

Positions were converted to distances from a fixed baseline with the short script below.

```
import numpy as np

def retreat(before, after):
    # positions in metres from baseline
    return float(np.mean(after - before))
```

3 Results

Table 1 summarises the survey. The longest retreat occurred at Long Shingle, where the beach lost 14.3 m in a single month.

Retreat was largest between November and February at every site. In the five summer months the shoreline moved by less than 0.4 m in total.

Harbour Wall, protected by a masonry wall, retreated least. All other sites lost between 2.3 m and 6.8 m on average.

Table 1. Shoreline retreat by site, October 2024 to September 2026.

Site	Mean retreat (m)	Maximum (m)	Storm events
North Spit	4.2	9.8	6
Harbour Wall	1.1	2.7	6
Long Shingle	6.8	14.3	7
Mill Point	3.5	8.1	5
South Cove	2.3	5.0	7

4 Discussion

The strong link between storm count and retreat suggests that monthly surveys are the minimum needed to manage these beaches. Sites with a hard defence behind the beach lost material more slowly, but they also had a narrower beach to begin with.

Our results are limited to five sites and two winters. Longer records are needed to separate storm effects from sea-level trends.

5 Conclusion

Gravel beaches on this coast retreat in short bursts, not steadily. A monthly survey is cheap, repeatable and sufficient to follow these bursts.

We recommend extending the programme to the remaining eleven beaches and adding a wave gauge at Long Shingle.

Acknowledgements

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References

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